

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent

12411275

Kind Code

B2

Date of Patent

September 09, 2025

Inventor(s)

Haag; Adam D. et al.

Optical stack having absorbing polarizer, half-wave retarder, and reflective polarizer

Abstract

An optical stack includes a reflective polarizer, an absorbing polarizer, and a half-wave retarder. The half-wave retarder is disposed between the reflective and absorbing polarizers. For substantially normally incident light and a first polarization state, the reflective polarizer reflects at least about 60% of the incident light for at least a first wavelength less than a cut-off wavelength and transmits at least about 50% of the incident light for at least a second wavelength greater than the cut-off wavelength. For substantially normally incident light, the half-wave retarder has a first retardance of less than about 250 nanometers (nm) at the first wavelength and a second retardance at the second wavelength. A deviation of the first retardance from a first half-wave retardance corresponding to the first wavelength is less than a deviation of the second retardance from a second half-wave retardance corresponding to the second wavelength.

Inventors: Haag; Adam D. (Woodbury, MN), Park; Sun-Yong (Gyeonggi-do, KR), Denker; Martin E. (Vadnais Heights, MN)

Applicant: 3M INNOVATIVE PROPERTIES COMPANY (St. Paul, MN)

Family ID: 80352742

Assignee: 3M INNOVATIVE PROPERTIES COMPANY (St. Paul, MN)

Appl. No.: 18/041432

Filed (or PCT Filed): July 29, 2021

PCT No.: PCT/IB2021/056936

PCT Pub. No.: WO2022/043791

PCT Pub. Date: March 03, 2022

Prior Publication Data

Related U.S. Application Data

us-provisional-application US 63070486 20200826

Publication Classification

Int. Cl.: G02B5/30 (20060101); G02F1/1335 (20060101); G02F1/13363 (20060101); H10K50/80 (20230101)

U.S. Cl.:

CPC G02B5/3083 (20130101); G02B5/3041 (20130101); G02B5/305 (20130101); G02F1/133536 (20130101); G02F1/133638 (20210101); H10K50/868 (20230201);

Field of Classification Search

CPC: G02B (5/3041); G02B (5/305); G02B (5/3083); G02F (1/133536); G02F (1/133638); H10K (50/868)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
10234727	12/2018	Sakai	N/A	G02F 1/13363
2017/0052369	12/2016	Shimatani et al.	N/A	N/A
2018/0039131	12/2017	Szwarcman et al.	N/A	N/A
2019/0004227	12/2018	Park et al.	N/A	N/A
2020/0012027	12/2019	Jeon et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2000131683	12/1999	JP	N/A
2017068111	12/2016	JP	N/A
2007124315	12/2006	WO	N/A

OTHER PUBLICATIONS

International Search Report for PCT Application No. PCT/IB2021/056936, mailed on Nov. 1, 2021, 4 pages. cited by applicant

Primary Examiner: Dunning; Ryan S

Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application is a national stage filing under 35 U.S.C. 371 of PCT/IB2021/056936, filed Jul. 29, 2021, which claims the benefit of U.S. Application No. 63/070,486, filed Aug. 26, 2020, the disclosures of which are incorporated by reference in their entirety herein.

TECHNICAL FIELD

(1) The present disclosure relates generally to an optical stack, and in particular, to an optical stack for a backlight.

BACKGROUND

(2) In backlights, reflective polarizers are generally used to recycle light, and enhance the ultimate efficiency and brightness of a display. Further, absorbing polarizers or the like are used in conjunction with a liquid crystal module to polarize the light for appropriate modulation by the liquid crystal module. In some manufacturing processes, reflective polarizers in roll form may have a reflection axis (i.e., block axis or slow axis) in a transverse direction (i.e., a width direction of the roll). Absorbing polarizers in roll form may have an absorption axis along a length direction of the roll. In some cases, it may be desirable to align the reflection axis of the reflective polarizer and the absorption axis of the absorbing polarizer within an optical stack. In conventional manufacturing process, one or both of the rolls of polarizers are cut and rotated, adding manufacturing time and process expense.

SUMMARY

(3) In a first aspect, the present disclosure provides an optical stack. The optical stack includes a reflective polarizer, an absorbing polarizer, and a half-wave retarder. The reflective polarizer includes mutually orthogonal transmission and reflection axes. The half-wave retarder is disposed between the reflective and absorbing polarizers. For substantially normally incident light and a first polarization state, the reflective polarizer includes a transmission band with a cut-off wavelength. The reflective polarizer reflects at least about 60% of the incident light for at least a first wavelength less than the cut-off wavelength and transmits at least about 50% of the incident light for at least a second wavelength greater than the cut-off wavelength. For substantially normally incident light and an orthogonal second polarization state, the reflective polarizer transmits at least 70% of the incident light for each of the at least first and second wavelengths. For substantially normally incident light, the half-wave retarder has a first retardance of less than about 250 nanometers (nm) at the first wavelength and a second retardance at the second wavelength. A deviation of the first retardance from a first half-wave retardance corresponding to the first wavelength is less than a deviation of the second retardance from a second half-wave retardance corresponding to the second wavelength. For substantially normally incident light and the first polarization state, the absorbing polarizer has a first transmittance at the first wavelength and a greater second transmittance at the second wavelength.

(4) In a second aspect, the present disclosure provides another optical stack. The optical stack includes a display panel, a reflective polarizer, an absorbing polarizer, and a half-wave retarder. The display panel is configured to emit a display light including a blue emission spectra with a blue full width at half maxima (FWHM) defined between a minimum wavelength and a maximum wavelength. The reflective polarizer is configured to receive and reflect a portion of the display light from the display panel as a reflected polarized light. The reflective polarizer includes mutually orthogonal transmission and reflection axes. The half-wave retarder is disposed between the reflective and absorbing polarizers. For substantially normally incident light and a first polarization state, the reflective polarizer includes a transmission band with a cut-off wavelength greater than the minimum wavelength of the blue emission spectra of the display panel. The reflective polarizer reflects at least about 60% of the incident light for at least a first wavelength less than the cut-off wavelength and transmits at least about 50% of the incident light for at least a second wavelength greater than the cut-off wavelength. For substantially normally incident light and an orthogonal

second polarization state, the reflective polarizer transmits at least 70% of the incident light for each of the at least first and second wavelengths. For substantially normally incident light, the half-wave retarder has a first retardance of less than about 250 nm at the first wavelength and a second retardance at the second wavelength. A deviation of the first retardance from a first half-wave retardance corresponding to the first wavelength is less than a deviation of the second retardance from a second half-wave retardance corresponding to the second wavelength. For substantially normally incident light and the first polarization state, the absorbing polarizer has a first transmittance at the first wavelength and a greater second transmittance at the second wavelength.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) Exemplary embodiments disclosed herein may be more completely understood in consideration of the following detailed description in connection with the following figures. The figures are not necessarily drawn to scale. Like numbers used in the figures refer to like components. However, it will be understood that the use of a number to refer to a component in a given figure is not intended to limit the component in another figure labeled with the same number.
- (2) FIG. 1 is a schematic exploded top perspective view of an optical stack according to one embodiment of the present disclosure;
- (3) FIG. 2 is a schematic exploded front elevation view of the optical stack of FIG. 1;
- (4) FIG. 3 is a schematic view of a half-wave retarder of the optical stack according to one embodiment of the present disclosure;
- (5) FIG. 4 is a detailed schematic view of the reflective polarizer according to one embodiment of the present disclosure;
- (6) FIG. 5 is a graph illustrating transmission versus wavelength for the reflective polarizer having a transmission band according to one embodiment of the present disclosure;
- (7) FIG. 6 is another graph illustrating transmission versus wavelength for the reflective polarizers having other transmission bands according to another embodiment of the present disclosure;
- (8) FIG. 7 is a schematic view of a half-wave retarder of the optical stack according to one embodiment of the present disclosure;
- (9) FIG. 8 is a graph illustrating retardation versus wavelength for the half-wave retarder according to one embodiment of the present disclosure;
- (10) FIG. 9 is a graph illustrating deviation of retardance versus wavelength for the half-wave retarder according to one embodiment of the present disclosure;
- (11) FIG. 10 is a schematic view of an absorbing polarizer of the optical stack according to one embodiment of the present disclosure; and
- (12) FIG. 11 is a graph illustrating transmission versus wavelength for the absorbing polarizer according to one embodiment of the present disclosure.

DETAILED DESCRIPTION

(13) In the following description, reference is made to the accompanying figures that form a part thereof and in which various embodiments are shown by way of illustration. It is to be understood that other embodiments are contemplated and may be made without departing from the scope or spirit of the present disclosure. The following detailed description, therefore, is not to be taken in a limiting sense.

(14) The present disclosure relates to an optical stack. The optical stack may be used in a backlight of electronic devices that incorporate displays, such as computer monitors, televisions, mobile phones, personal digital assistants (PDAs), wearable devices and other portable devices. A typical optical stack includes a reflective polarizer and an absorbing polarizer.

(15) In conventional manufacturing processes and with conventional material selections, a

reflective polarizer is stretched such that a reflection axis (the axis parallel to a polarization state preferentially reflected by the reflective polarizer) is along a width direction of the reflective polarizer. Further, a conventional absorbing polarizer is usually aligned in the machine direction, resulting in an absorption axis being substantially along a length of the absorbing polarizer. Thus, conventional manufacturing processes result in rolls of the reflective polarizer and the absorbing polarizer having transmission axes oriented orthogonally to one another. Therefore, costly converting steps may be needed in order to provide a sheet having an absorbing polarizer and a reflective polarizer with their transmission axes aligned. Generally, in a sheet to roll lamination process, the reflective polarizer is cut and rotated 90 degrees in order to align the transmission axes. This process is time consuming and expensive, and sometimes the converting steps may also increase the chance of introducing defects, which can lower the yield or usable portion of the optical stack. Therefore, it may be desirable to align the reflection axis of the reflective polarizer and the absorption axis of the absorbing polarizer within an optical stack to save additional manufacturing time and process expense.

(16) In some designs, a half-wave retarder is disposed between the reflective polarizer and the absorbing polarizer. The half-wave retarder may have a slow axis oriented substantially 45 degrees to a transmission axis of the reflective polarizer. The half-wave retarder may rotate linearly polarized light 90 degrees between the reflective polarizer and the absorbing polarizer to align the reflection axis of the reflective polarizer and the absorption axis of the absorbing polarizer. However, incorporation of the half-wave retarder may pose a challenge to rotate the linearly polarized light evenly for all visible wavelengths to minimize color shifts, or other artifacts at various viewing angles in the optical stack. The color shifts may be due to non-linear wavelength-dependent modulation of light in a conventional half-wave retarder. In some of these cases, specialized materials are used to make achromatic half-wave retarders to reduce the color shifts. However, even an achromatic half-wave retarder may display some chromatic differences in conversion of linearly polarized light. Further, utilizing achromatic half-wave retarders may also significantly increase the overall cost of the optical stack.

(17) The present disclosure relates to an optical stack including a reflective polarizer, an absorbing polarizer, and a half-wave retarder. The reflective polarizer of the present disclosure has mutually orthogonal transmission and reflection axes. The half-wave retarder is disposed between the reflective and absorbing polarizers. For substantially normally incident light and a first polarization state, the reflective polarizer of the present disclosure includes a transmission band with a cut-off wavelength. For the first polarization state, the reflective polarizer reflects at least about 60% of the incident light for at least a first wavelength less than the cut-off wavelength, and transmits at least about 50% of the incident light for at least a second wavelength greater than the cut-off wavelength. For substantially normally incident light and an orthogonal second polarization state, the reflective polarizer transmits at least 70% of the incident light for each of the at least first and second wavelengths.

(18) Furthermore, the half-wave retarder of the present disclosure is designed in such a way that, for substantially normally incident light, the half-wave retarder has a first retardance of less than about 250 nanometers (nm) at the first wavelength and a second retardance at the second wavelength. A deviation of the first retardance from a first half-wave retardance corresponding to the first wavelength is less than a deviation of the second retardance from a second half-wave retardance corresponding to the second wavelength.

(19) Such optimization of the half-wave retarder in the wavelengths less than the cut-off wavelength of the transmission band of the reflective polarizer may reduce optical artifacts, such as color shifts. Specifically, the combination of the reflective polarizer and the half-wave retarder of the present disclosure may exhibit an improved performance in reducing the color shifts. Further, a half-wave retarder optimized in such a manner may be easier and cheaper to produce as compared to conventional achromatic half-wave retarders.

(20) Referring now to FIGS. 1 and 2, an optical stack **100** is illustrated according to an embodiment of the present disclosure. The optical stack **100** includes a reflective polarizer **102**, a half-wave retarder **104**, an absorbing polarizer **106**, and a display panel **108**. In some embodiments, the reflective polarizer **102**, the half-wave retarder **104**, the absorbing polarizer **106**, and the display panel **108** are substantially co-extensive with each other, or of same in-plane dimensions (i.e., length and width).

(21) The optical stack **100** defines mutually orthogonal x, y, and z-axes. The x and y-axes are in-plane axes of the optical stack **100**, while the z-axis is a transverse axis disposed along a thickness of the optical stack **100**. In other words, the x and y-axes are disposed along a plane of the optical stack **100**, while the z-axis is perpendicular to the plane of the optical stack **100**. The reflective polarizer **102**, the half-wave retarder **104**, the absorbing polarizer **106**, and the display panel **108** are disposed adjacent to each other along the z-axis of the optical stack **100**.

(22) In some embodiments, the reflective polarizer **102**, the half-wave retarder **104**, and the absorbing polarizer **106** of the optical stack **100** may be in optical contact with one another. In some embodiments, the reflective polarizer **102**, the half-wave retarder **104**, and the absorbing polarizer **106** of the optical stack **100** are attached with one or more of a pressure sensitive adhesive, an optically clear adhesive, an Ultraviolet (UV) curable adhesive, or a polyvinyl alcohol type adhesive, laminations, or any other attachment mechanisms.

(23) The reflective polarizer **102** includes mutually orthogonal transmission and reflection axes **122**, **124**. The reflective polarizer **102** may be a notched reflective polarizer having one or more notches or bands. The reflective polarizer **102** may be any suitable reflective polarizer, including a wire-grid polarizer or a multilayer birefringent reflective polarizer. In the illustrated embodiment, the reflective polarizer **102** may have the transmission axis **122** generally along the x-axis of the reflective polarizer and the reflection axis **124** generally along the y-axis. The reflective polarizer **102** further includes a first major surface **112** and a second major surface **114** opposite to the first major surface **112**. In some embodiments, each of the first and second major surfaces **112**, **114** of the reflective polarizer **102** may be substantially planar and disposed along the x-y plane.

(24) The half-wave retarder **104** is disposed between the reflective and absorbing polarizers **102**, **106**. The half-wave retarder **104** includes a first major surface **116** and a second major surface **118** opposite to the first major surface **116**. In some embodiments, each of the first and second major surfaces **116**, **118** of the half-wave retarder **104** may be substantially planar and disposed along the x-y plane. In the illustrated embodiment of FIG. 2, the absorbing polarizer **106** is disposed adjacent to the first major surface **116** of the half-wave retarder **104**. Further, the second major surface **118** of the half-wave retarder **104** is disposed adjacent to the first major surface **112** the reflective polarizer **102**. Generally, the half-wave retarder **104** includes a layer of a birefringent material. In some embodiments, the half-wave retarder **104** includes a half-wave cyclo-olefin polymer (COP) retarder. In some embodiments, the half-wave retarder **104** may include two layers of birefringent material. In some embodiments, the half-wave retarder **104** may be a blend of two materials, i.e., a positive birefringent material and a negative birefringent material. In some embodiments, the half-wave retarder **104** may include two layers such that one of the layers may rotate linearly polarized light by substantially 15 degrees, and another layer may rotate linearly polarized light by substantially 75 degrees. In some embodiments, the half-wave retarder **104** is a non-achromatic half-wave retarder, i.e., a retardance of the half-wave retarder **104** substantially deviates from an ideal half-wave retardance across at least a portion of a visible wavelength range. Generally, a deviation of the retardance of the achromatic half-wave retarder from the retardance of an ideal half-wave retarder is less than a deviation of the retardance of the non-achromatic retarder from the retardance of the ideal half-wave retarder. Further, the retardance of the non-achromatic half-wave retarder is generally same or less at higher wavelengths as compared to the retardance at lower wavelengths. On the other hand, the retardance of the achromatic half-wave retarder generally increases at the higher wavelengths.

(25) The half-wave retarder **104** includes a slow axis **126** oriented substantially 45 degrees with respect to the transmission axis **122** of the reflective polarizer **102**. For the purposes of this application, an orientation of substantially 45 degrees and substantially 135 degrees may be considered substantially equivalent, given the bidirectional nature of the transmission axis **122**. Substantially 45 degrees also may be understood to not be limited to precisely 45 degrees, instead, the alignment of the axes may be within 10 degrees, within 5 degrees, or within 1 degrees of 45 degrees. Alignment may in some cases be a tradeoff between manufacturability (e.g., error tolerance) and optical performance, the appropriate balance being determined depending on the application attributes.

(26) The display panel **108** is disposed below the reflective polarizer **102** opposite to the half-wave retarder **104**. In the illustrated embodiment of FIG. 2, the display panel **108** is disposed adjacent to the second major surface **114** of the reflective polarizer **102**. In some embodiments, the display panel **108** includes an organic light emitting diode (OLED) display panel. In some embodiments, a quarter-wave retarder (not shown) may be disposed between the reflective polarizer **102** and the display panel **108**. The quarter-wave retarder may be used to produce circular polarization in order to reduce ambient reflection from the display panel **108**. In some other embodiments, the display panel **108** includes a liquid crystal display (LCD) panel.

(27) The absorbing polarizer **106** may be disposed on the half-wave retarder **104** opposite to the reflective polarizer **102**. In the illustrated embodiment of FIG. 2, a bottom surface **120** of the absorbing polarizer **106** is disposed on the first major surface **116** of the half-wave retarder **104**. The absorbing polarizer **106** includes a transmission axis **128** oriented substantially 90 degrees with respect to the transmission axis **122** of the reflective polarizer **102**, and substantially 45 degrees (or 135 degrees) with respect to the slow axis **126** of the half-wave retarder **104**. In the illustrated embodiment, the transmission axis **128** of the absorbing polarizer **106** is disposed generally along the y-axis. The absorbing polarizer **106** may include any suitable material, such as a polymeric material. In some embodiments, the absorbing polarizer **106** may include polyvinyl alcohol. In some embodiments, the absorbing polarizer **106** may include polarizing elements, including polarizing or dichroic dyes.

(28) In some embodiments, the optical stack **100** may include additional or intermediate films, layers, or components, such as, diffusing layers, turning layers, or substrate layers. The optical stack **100** may, in total, be of any suitable thickness based on desired application attributes.

(29) The display panel **108** is configured to emit a display light **10**. The reflective polarizer **102** is configured to receive and reflect a portion of the display light **10** from the display panel **108** as a reflected polarized light **12**. The display light **10** is generally unpolarized. However, in some cases, the display light **10** may be at least partially polarized light. For the purpose of explanation, the display light **10** may be treated as light having an unknown or arbitrary polarization state or distribution of polarization states. The reflected polarized light **12** is generally polarized along the reflection axis **124** (i.e., y-axis) of the reflective polarizer **102**. The reflected polarized light **12** may therefore have a first polarization state along the y-axis. The reflective polarizer **102** transmits a first transmitted light **14** having an orthogonal second polarization state parallel to the transmission axis **122** (i.e., x-axis) of the reflective polarizer **102**.

(30) The half-wave retarder **104** receives the first transmitted light **14** from the reflective polarizer **102**. The half-wave retarder **104** is configured to rotate or modulate at least a portion of the polarization of first transmitted polarized light **14** to orthogonally polarized second transmitted light **16**. The second transmitted light **16** may therefore have the first polarization state. The absorbing polarizer **106** receives the second transmitted light **16** from the half-wave retarder **104** and emits at least a portion of the second transmitted light **16** as an output light **18** having the first polarization state.

(31) FIGS. 3 and 4 illustrate the reflective polarizer **102**. The reflective polarizer **102** may include one or more of a polymeric reflective polarizer, a wire grid reflective polarizer, and a diffuse

reflective polarizer. As illustrated in FIG. 4, the reflective polarizer **102** includes a plurality of alternating first and second polymeric layers **302**, **304**. In some embodiments, the plurality of alternating first and second polymeric layers **302**, **304** number at least 40 in total. In some other embodiments, the plurality of alternating first and second polymeric layers **302**, **304** number at least 50, at least 75, at least 90, at least 95, or at least 100 in total. Each of the first and second polymeric layers **302**, **304** defines an average thickness “T” along the z-axis. In some embodiments, each of the first and second polymeric layers **302**, **304** has the average thickness “T” of less than about 350 nm. In some embodiments, each of the first and second polymeric layers **302**, **304** may have the average thickness “T” less than about 400 nm. In some other embodiments, the average thickness “T” of each of the first and second polymeric layers **302**, **304** may be less than about 500 nm, less than about 600, or less than about 700 nm. In some embodiments, the reflective polarizer **102** may further include a protective layer **306** disposed on each major side of the plurality of alternating first and second polymeric layers **302**, **304**.

(32) FIGS. 5 and 6 illustrate exemplary graphs **400**, **500**, respectively. The graphs **400**, **500** depict transmission percentage versus wavelength for different transmission bands of the reflective polarizer **102**. Wavelength is expressed in nanometers (nm) across the visible wavelength range. In FIGS. 5 and 6, transmittance is expressed as transmission percentage in the left ordinate. In FIG. 5, reflectance is expressed as reflection percentage in the right ordinate. The reflection percentage is complementary to the transmission percentage, i.e., reflection percentage=(100–transmission percentage).

(33) Referring to FIGS. 1, 5 and 6, the display panel **108** is configured to emit the display light **10** including a blue emission spectra **402** with a blue full width at half maxima (FWHM) **410** defined between a minimum wavelength **406** and a maximum wavelength **407**. Any suitable energy unit may be used to illustrate the blue emission spectra **402** of the display panel **108** in the graphs **400**, **500**. In some embodiments, the minimum wavelength **406** is greater than about 430 nm and the maximum wavelength **407** is less than about 480 nm. In some other embodiments, the minimum wavelength **406** may be greater than about 400 nm and the maximum wavelength **407** may be less than about 500 nm.

(34) Referring to FIGS. 3, 4 and 5, for substantially normally incident light **202**, and the first polarization state, the reflective polarizer **102** includes a transmission band **404** with a cut-off wavelength **408**. Specifically, for the substantially normally incident light **202** and the first polarization state, the reflective polarizer **102** includes a transmission spectrum **403** including the transmission band **404** with the cut-off wavelength **408**. The minimum wavelength **406** is less than the cut-off wavelength **408** of the transmission band **404** of the reflective polarizer **102**. In other words, the reflective polarizer **102** includes the transmission band **404** with the cut-off wavelength **408** greater than the minimum wavelength **406** of the blue emission spectra **402** of the display panel **108**. In the illustrated embodiment of FIG. 5, the maximum wavelength **407** is also less than the cut-off wavelength **408** of the transmission band **404** of the reflective polarizer **102**. In other words, the cut-off wavelength **408** of the transmission band **404** of the reflective polarizer **102** is greater than the maximum wavelength **407** of the blue FWHM **410** of the display panel **108**.

(35) The cut-off wavelength **408** may correspond to a wavelength at which transmittance is half (50%) of a peak or a maximum transmittance. The cut-off wavelength **408** of the transmission band **404** of the reflective polarizer **102** is greater than about 500 nm. In some other embodiments, the cut-off wavelength **408** of the transmission band **404** of the reflective polarizer **102** is greater than about 525 nm. In some other embodiments, the cut-off wavelength **408** of the transmission band **404** of the reflective polarizer **102** is less than about 550 nm.

(36) For the first polarization state, the reflective polarizer **102** reflects at least about 60% of the incident light **202** for at least a first wavelength W1 less than the cut-off wavelength **408** and transmits at least about 50% of the incident light **202** for at least a second wavelength W2 greater than the cut-off wavelength **408**. In some other embodiments, for the first polarization state, the

reflective polarizer **102** reflects at least about 70%, at least about 80%, at least about 90%, at least about 95%, or at least about 99% of the incident light **202** for at least the first wavelength **W1** less than the cut-off wavelength **408**. In some embodiments, for the first polarization state, the reflective polarizer **102** transmits at least about 55%, at least about 60%, at least about 70%, at least about 80%, or at least about 90% of the incident light **202** for at least the second wavelength **W2** greater than the cut-off wavelength **408**. The first wavelength **W1** lies in a blue wavelength range. Therefore, for the first polarization state, the reflective polarizer **102** reflects at least a portion of the wavelengths in the blue wavelength range. In some embodiments, the first wavelength **W1** lies in a range from about 400 nm to about 500 nm. In the illustrated embodiment, the first wavelength **W1** is less than the minimum wavelength **406**. In some other embodiments, the first wavelength **W1** may be between the minimum wavelength **406** and the maximum wavelength **407**. The second wavelength **W2** may lie in a red wavelength range. In some embodiments, the second wavelength **W2** lies in a range from about 600 nm to about 700 nm. In some embodiments, for the first polarization state, the reflective polarizer **102** may reflect at least about 60% of the incident light **202** for one or more wavelengths in the blue wavelength range and may transmit at least about 50% of the incident light **202** for each wavelength in subsequent wavelength ranges, such as a green wavelength range and the red wavelength range.

(37) For the substantially normally incident light **202**, and the second polarization state, the reflective polarizer **102** transmits at least 70% of the incident light **202** for each of the at least first and second wavelengths **W1**, **W2**. In some embodiments, for the second polarization state, the reflective polarizer **102** transmits at least about 80%, at least about 85%, or at least about 90% of the incident light **202** for each of the at least first and second wavelengths **W1**, **W2**. As shown in FIG. 5, for the substantially normally incident light **202**, and the second polarization state, the reflective polarizer **102** includes a transmission spectrum **420**. In some embodiments, for the second polarization state, the reflective polarizer **102** may transmit at least 70% of the incident light **202** for each wavelength in a visible wavelength range from about 400 nm to about 700 nm. In some other embodiments, for the second polarization state, the reflective polarizer **102** may transmit at least 80%, at least 85%, or at least 90% of the incident light **202** for each wavelength in the visible wavelength range.

(38) In some embodiments, the first polarization state is S polarization state and the second polarization state is P polarization state. In some other embodiments, the first polarization state is P polarization state and the second polarization state is S polarization state. The first polarization state is generally along the y-axis, while the second polarization state is generally along the x-axis.

(39) Referring to FIGS. 3, 4 and 6, for the substantially normally incident light **202**, and the first polarization state, the reflective polarizer **102** includes any one of transmission bands **502**, **506**, **510**. The transmission bands **502**, **506**, **510** correspond to transmission spectra **522**, **524**, **526**, respectively, for the substantially normally incident light **202**, and the first polarization state. The transmission spectrum **420** (shown in FIG. 5) for the substantially normally incident light **202** and the second polarization state may correspond to each of the transmission spectra **522**, **524**, **526** for the first polarization state.

(40) The transmission band **502** has a cut-off wavelength **504**. The transmission band **506** has a cut-off wavelength **508**. The transmission band **510** has a cut-off wavelength **512**. The minimum wavelength **406** is less than each of the cut-off wavelengths **504**, **508**, **512** of the respective transmission bands **502**, **506**, **510**. In other words, the cut-off wavelengths **504**, **508**, **512** of the respective transmission bands **502**, **506**, **510** are greater than the minimum wavelength **406** of the blue FWHM **410** of the display panel **108**. In the illustrated embodiment of FIG. 6, the maximum wavelength **407** is less than each of the cut-off wavelengths **508**, **512** of the respective transmission bands **506**, **510**. In other words, the cut-off wavelengths **508**, **512** of the transmission bands **506**, **510** are greater than the maximum wavelength **407** of the blue FWHM **410** of the display panel **108**. However, the cut-off wavelength **504** of the transmission band **502** is less than the maximum

wavelength **407** of the blue FWHM **410** of the display panel **108**. The cut-off wavelength **504** of the transmission band **502** is therefore greater than the minimum wavelength **406** and less than the maximum wavelength **407**. In some embodiments, the cut-off wavelength **504** of the transmission band **502** of the reflective polarizer **102** is about 450 nm. In some embodiments, the cut-off wavelength **504** of the transmission band **502** of the reflective polarizer **102** is greater than about 425 nm and less than about 475 nm. In some other embodiments, the cut-off wavelength **504** of the transmission band **502** of the reflective polarizer **102** is greater than about 470 nm. In some embodiments, the cut-off wavelength **508** of the transmission band **506** of the reflective polarizer **102** is about 525 nm. In some embodiments, the cut-off wavelength **508** of the transmission band **506** of the reflective polarizer **102** is greater than about 500 nm and less than about 550 nm. In some embodiments, the cut-off wavelength **512** of the transmission band **510** of the reflective polarizer **102** is about 475 nm. In some other embodiments, the cut-off wavelength **512** of the transmission band **510** of the reflective polarizer **102** is greater than about 450 nm and less than about 500 nm.

(41) For the first polarization state, the reflective polarizer **102** reflects at least about 60% of the incident light **202** for at least the first wavelength **W1** less than each of the cut-off wavelengths **504**, **508**, **512** and transmits at least about 50% of the incident light **202** for at least the second wavelength **W2** greater than each of the cut-off wavelengths **504**, **508**, **512**. In some other embodiments, for the first polarization state, the reflective polarizer **102** reflects at least about 70%, at least about 80%, at least about 90%, at least about 95%, or at least about 99% of the incident light **202** for at least the first wavelength **W1** less than each of the cut-off wavelengths **504**, **508**, **512**. In some embodiments, for the first polarization state, the reflective polarizer **102** transmits at least about 50%, at least about 60%, at least about 70%, at least about 80%, or at least about 90% of the incident light **202** for at least the second wavelength **W2** greater than the each of the cut-off wavelengths **504**, **508**, **512**. The first wavelength **W1** lies in the blue wavelength range. Therefore, for the first polarization state, the reflective polarizer **102** reflects a portion of the wavelengths in the blue wavelength range.

(42) For substantially normally incident light **202**, and the second polarization state, the reflective polarizer **102** transmits at least 70% of the incident light **202** for each of the at least first and second wavelengths **W1**, **W2**. In some other embodiments, for the second polarization state, the reflective polarizer **102** transmits at least about 80%, at least about 85%, or at least about 90% of the incident light **202** for each of the at least first and second wavelengths **W1**, **W2**.

(43) FIG. 7 illustrates the half-wave retarder **104**. FIG. 8 is a graph **600** illustrating retardation versus wavelength for the half-wave retarder **104** according to one embodiment of the present disclosure. The graph **600** includes a retardation curve **602** which depicts retardation of the half-wave retarder **104** at various wavelengths.

(44) Referring to FIGS. 7 and 8, for substantially normally incident light **204**, the half-wave retarder **104** has a first retardance **R1** of less than about 250 nm at the first wavelength **W1** and a second retardance **R2** at the second wavelength **W2**. In some embodiments, the second retardance **R2** is greater than about 200 nm. In some embodiments, the second retardance **R2** is less than the first retardance **R1**. In other words, the second retardance **R2** is less than about 250 nm.

(45) Furthermore, as shown in FIG. 8, the half-wave retarder **104** has a minimum blue retardance **Rb** at a corresponding blue wavelength **Wb**, a minimum green retardance **Rg** at a corresponding green wavelength **Wg**, and a minimum red retardance **Rr** at a corresponding red wavelength **Wr**. The minimum blue, green, and red retardances **Rb**, **Rg**, **Rr** are within 20% of range of each other. In some other embodiments, the minimum blue, green, and red retardances **Rb**, **Rg**, **Rr** are within 15% of range of each other. As illustrated in FIG. 8, the minimum blue, green, and red retardances **Rb**, **Rg**, **Rr** are less than 250 nm. In some embodiments, the second retardance **R2** is less than the minimum blue and green retardances **Rb**, **Rg**. In some embodiments, the second retardance **R2** is greater than the minimum red retardance **Rr**. In some embodiments, the first retardance **R1** is

greater than the minimum blue, green, and red retardances Rb, Rg, Rr. It may be apparent from the graph **600** that the retardance of the half-wave retarder **104** does not vary considerably with the wavelength of the incident light **204**. The half-wave retarder **104** is therefore a non-achromatic half-wave retarder.

(46) Referring to FIG. **8**, a half-wave curve **604** denotes a variation of retardance with wavelength for an ideal half-wave retarder. In other words, a retardance at each point on the half-wave curve **604** is half of the corresponding wavelength (i.e., $\text{Retardance} = \text{Wavelength}/2$). The half-wave curve **604** intersects the retardation curve **602** of the half-wave retarder **104** at only one point C. Therefore, the half-wave retarder **104** has an ideal half-wave retardance at only one wavelength W_c corresponding to the point C. The wavelength W_c may be about 450 nm. Otherwise, the retardance curve **602** deviates from the half-wave curve **604** at all other wavelengths. Further, the retardance curve **602** generally decreases with an increase in wavelength, while the half-wave curve **604** increases with an increase in wavelength. The half-wave curve **604** has a first half-wave retardance H_1 at the first wavelength W_1 (i.e., $H_1 = W_1/2$). Further, the half-wave curve **604** has a second half-wave retardance H_2 at the second wavelength W_2 (i.e., $H_2 = W_2/2$).

(47) FIG. **9** is a graph **700** illustrating deviation of retardance versus wavelength for the half-wave retarder **104**. The graph **700** includes a retardance deviation curve **702** which depicts deviation of retardance of the half-wave retarder **104** at various wavelengths.

(48) Referring to FIGS. **8** and **9**, the retardance deviation curve **702** illustrates a deviation of the retardance curve **602** of the half-wave retarder **104** from the half-wave curve **604**. A deviation D_1 of the first retardance R_1 from the first half-wave retardance H_1 corresponding to the first wavelength W_1 is less than a deviation D_2 of the second retardance R_2 from the second half-wave retardance H_2 corresponding to the second wavelength W_2 . The first half-wave retardance H_1 corresponding to the first wavelength W_1 may be considered as an ideal retardance of the half-wave retarder **104** at the first wavelength W_1 . Similarly, the second half-wave retardance H_2 corresponding to the second wavelength W_2 may be considered as an ideal retardance of the half-wave retarder **104** at the second wavelength W_2 . It may be apparent from the graph **700** that a deviation of the retardance of the half-wave retarder **104** from the half-wave curve **604** for a wavelength range from about 400 nm to about 500 nm is lower than a deviation for wavelengths greater than 500 nm (for example, in the red wavelength range). The wavelength range from about 400 nm to about 500 nm may correspond to the blue wavelength range. Therefore, the half-wave retarder **104** may be optimized for the blue wavelength range. In other words, the half-wave retarder **104** may behave closer to an ideal half-wave retarder within the wavelength range from about 400 nm to about 500 nm as compared to other wavelengths.

(49) As described above with respect to FIGS. **3**, **5** and **6**, for substantially normally incident light **202** and the first polarization state, the reflective polarizer **102** reflects at least about 60% of the incident light **202** for at least the first wavelength W_1 less than each of the cut-off wavelengths **408**, **504**, **508**, **512** of the respective transmission bands **404**, **502**, **506**, **510**, and transmits at least about 50% of the incident light **202** for at least the second wavelength W_2 greater than each of the cut-off wavelengths **408**, **504**, **508**, **512** of the respective transmission bands **404**, **502**, **506**, **510**. For substantially normally incident light **204**, the half-wave retarder **104** has the first retardance R_1 of less than about 250 nm at the first wavelength W_1 and the second retardance R_2 at the second wavelength W_2 . The second retardance R_2 is less than the first retardance R_1 . Further, the deviation D_1 of the first retardance R_1 from the first half-wave retardance H_1 corresponding to the first wavelength W_1 is less than the deviation D_2 of the second retardance R_2 from the second half-wave retardance H_2 corresponding to the second wavelength W_2 .

(50) Such optimization of the half-wave retarder **104** in the wavelengths less than the cut-off wavelengths **408**, **504**, **508**, **512** of the respective transmission bands **404**, **502**, **506**, **510** may reduce color shifts during operation of the optical stack **100** of FIGS. **1** and **2**. Specifically, using the half-wave retarder **104** optimized in the blue wavelength range with the reflective polarizer **102**

having the cut-off wavelengths **408, 504, 508, 512** of the respective transmission bands **404, 502, 506, 510** in or near the blue wavelength range may reduce the color shifts. The combination of the reflective polarizer **102** and the half-wave retarder **104** may therefore exhibit an improved performance in reducing the color shifts. Further, a half-wave retarder optimized in such a manner may be easier and cheaper to manufacture as compared to conventional achromatic half-wave retarders having multiple layers. It may be noted that any one of the possible transmission bands **404, 502, 506, 510** of the reflective polarizer **102** may be chosen based on desired application attributes, such as the retardance characteristics of the half-wave retarder **104**, emission properties of the display panel **108**, and so forth.

(51) FIG. **10** illustrates the absorbing polarizer **106**. FIG. **11** is a graph **800** illustrating transmission percentage versus wavelength for the absorbing polarizer **106**. The graph **800** includes a transmission curve **802** which depicts transmission percentage of the absorbing polarizer **106** at various wavelengths. Referring to FIGS. **10** and **11**, for substantially normally incident light **206** and the first polarization state, the absorbing polarizer **106** has a first transmittance **T1** at the first wavelength **W1** and a greater second transmittance **T2** at the second wavelength **W2**. In some embodiments, for the first polarization state, the first transmittance **T1** of the absorbing polarizer **106** is at least about 60%. In some embodiments, for the first polarization state, the first transmittance **T1** of the absorbing polarizer **106** may be at least about 70%. In some embodiments, for the first polarization state, the second transmittance **T2** of the absorbing polarizer **106** is at least about 65%. In some other embodiments, for the first polarization state, the second transmittance **T2** of the absorbing polarizer **106** may be at least about 75%, at least about 80%, at least about 85%, or at least about 90%. In some embodiments, for the substantially normally incident light **206** and the second polarization state, the absorbing polarizer **106** may absorb at least 70% of the incident light **206** in the visible wavelength range.

(52) The absorbing polarizer **106** may therefore transmit less light in the blue wavelength range as compared to other wavelengths. The absorbing polarizer **106** when combined with the reflective polarizer **102** and the half-wave retarder **104** in the optical stack **100** of FIGS. **1** and **2** may further reduce the color shifts.

(53) Unless otherwise indicated, all numbers expressing feature sizes, amounts, and physical properties used in the specification and claims are to be understood as being modified by the term “about”. Accordingly, unless indicated to the contrary, the numerical parameters set forth in the foregoing specification and attached claims are approximations that can vary depending upon the desired properties sought to be obtained by those skilled in the art utilizing the teachings disclosed herein.

(54) Although specific embodiments have been illustrated and described herein, it will be appreciated by those of ordinary skill in the art that a variety of alternate and/or equivalent implementations can be substituted for the specific embodiments shown and described without departing from the scope of the present disclosure. This application is intended to cover any adaptations or variations of the specific embodiments discussed herein. Therefore, it is intended that this disclosure be limited only by the claims and the equivalents thereof.

Claims

1. An optical stack comprising: a reflective polarizer comprising mutually orthogonal transmission and reflection axes; an absorbing polarizer; and a half-wave retarder disposed between the reflective and absorbing polarizers, such that for substantially normally incident light: for a first polarization state, the reflective polarizer comprises a transmission band with a cut-off wavelength, wherein the reflective polarizer reflects at least about 60% of the incident light for at least a first wavelength less than the cut-off wavelength and transmits at least about 50% of the incident light for at least a second wavelength greater than the cut-off wavelength; for an orthogonal second

polarization state, the reflective polarizer transmits at least 70% of the incident light for each of the at least first and second wavelengths; the half-wave retarder has a first retardance of less than about 250 nanometers (nm) at the first wavelength and a second retardance at the second wavelength, wherein a deviation of the first retardance from a first half-wave retardance corresponding to the first wavelength is less than a deviation of the second retardance from a second half-wave retardance corresponding to the second wavelength; and for the first polarization state, the absorbing polarizer has a first transmittance at the first wavelength and a greater second transmittance at the second wavelength.

2. The optical stack of claim 1, wherein the cut-off wavelength of the transmission band of the reflective polarizer is less than about 550 nm and greater than about 450 nm.

3. The optical stack of claim 1, wherein the reflective polarizer comprises a plurality of alternating first and second polymeric layers numbering at least 40 in total, each of the first and second polymeric layers having an average thickness of less than about 350 nm.

4. The optical stack of claim 1, wherein the second retardance is greater than about 200 nm, and wherein the first transmittance of the absorbing polarizer is at least about 60%.

5. The optical stack of claim 1, further comprising a display panel disposed below the reflective polarizer opposite to the half-wave retarder, the display panel configured to emit a display light including a blue emission spectra with a blue full width at half maxima (FWHM) defined between a minimum wavelength and a maximum wavelength, wherein the minimum wavelength is less than the cut-off wavelength of the transmission band of the reflective polarizer.

6. The optical stack of claim 5, wherein the cut-off wavelength of the transmission band of the reflective polarizer is less than the maximum wavelength of the blue FWHM.

7. The optical stack of claim 5, wherein the cut-off wavelength of the transmission band of the reflective polarizer is greater than the maximum wavelength of the blue FWHM.

8. An optical stack comprising: a display panel configured to emit a display light including a blue emission spectra with a blue full width at half maxima (FWHM) defined between a minimum wavelength and a maximum wavelength; a reflective polarizer configured to receive and reflect a portion of the display light from the display panel as a reflected polarized light, the reflective polarizer comprising mutually orthogonal transmission and reflection axes; an absorbing polarizer; a half-wave retarder disposed between the reflective and absorbing polarizers, such that for substantially normally incident light; for a first polarization state, the reflective polarizer comprises a transmission band with a cut-off wavelength greater than the minimum wavelength of the blue emission spectra of the display panel, wherein the reflective polarizer reflects at least about 60% of the incident light for at least a first wavelength less than the cut-off wavelength and transmits at least about 50% of the incident light for at least a second wavelength greater than the cut-off wavelength; for an orthogonal second polarization state, the reflective polarizer transmits at least 70% of the incident light for each of the at least first and second wavelengths; the half-wave retarder has a first retardance of less than about 250 nm at the first wavelength and a second retardance at the second wavelength, wherein a deviation of the first retardance from a first half-wave retardance corresponding to the first wavelength is less than a deviation of the second retardance from a second half-wave retardance corresponding to the second wavelength; and for the first polarization state, the absorbing polarizer has a first transmittance at the first wavelength and a greater second transmittance at the second wavelength.

9. The optical stack of claim 8, wherein the minimum wavelength is greater than about 430 nm and the maximum wavelength is less than about 480 nm, and wherein the cut-off wavelength of the transmission band of the reflective polarizer is less than the maximum wavelength of the blue FWHM.

10. The optical stack of claim 8, wherein the cut-off wavelength of the transmission band of the reflective polarizer is greater than the maximum wavelength of the blue FWHM.
